Empirical Legal Evaluation on Harvey Legal Agent Benchmark (LAB): Controlled Direct LLM Tool-Calling (Study A) and Autonomous Agent Harness Integration (Study B)

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Benchmark & Harness Context: harveyai/harvey-labs (b58a28a) and google-research/harness2 (arXiv:2610.14224)
Scope: 4 multi-document legal tasks, 20 source commercial documents, 199 official binary rubric criteria

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1. Two Complementary Evaluation Protocols (Section 9.1)

Evaluating an evidence-structuring layer inside an autonomous coding agent (OpenCode with bash and filesystem access) confounds representation quality with agent shell exploration, prompt length (9,456 vs. 2,333 characters), and file-packaging repairs. Conversely, evaluating an evidence layer only on short synthetic QA misses complex multi-document legal deliverables.

To address both boundaries, we report two paired evaluations on the same four Harvey LAB tasks (199 criteria):

1. **Study A — Controlled Direct LLM Tool-Calling (`direct_rag` vs. `direct_claimlayer`, No Agent Harness, No `bash`):**
   - **Identical Solver Model:** nvidia/nemotron-3-super-120b-a12b (temperature=0.0, enable_thinking=False).
   - **Identical System Prompt & Tool Budget:** Both arms receive the exact same 650-character system prompt, no bash or filesystem shell, and an identical budget of **5 tool-calling rounds** followed by **1 final synthesis call**.
   - **Identical Container Packaging:** Both arms convert their final Markdown report to .docx via pandoc with automated ZIP header (PK\x03\x04) verification prior to grading (zero container failures in either arm).
   - **Identical Single Judge Across All 4 Tasks:** evaluation.run_eval with nvidia/nemotron-3-super-120b-a12b.
2. **Study B — Exploratory Autonomous Agent Harness Integration (`OpenCode` / `Harness²` Component Tree `c5+c7`):**
   - Compares the base LAB/OpenCode (v1.14.18, tool_mode="base", skill_mode="native", openai/gpt-4.1) agent against an augmented 8-component harness (c1–c8) after document-container repair.

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2. Study A: Controlled Direct LLM Tool-Calling (No `bash`, No Harness Confounds)

2.1 Tool Surfaces Compared

- **Arm 1 (`direct_rag` — Standard Chunk Retrieval Tools):** Documents are parsed with the LAB parser and segmented into overlapping 1,200-character chunks (200-character overlap) indexed with BM25. The LLM has three tools:
    1. `list_documents()`
    2. `search_chunks(query, filename_filter, top_k)`
    3. `read_chunk_window(filename, start_chunk, num_chunks)`
- **Arm 2 (`direct_claimlayer` — ClaimLayer / Deltoryx Structured Evidence & Diff Tools):** All paragraphs across all case documents are indexed into ClaimLayer/Deltoryx claim units (with no 80-paragraph truncation), tagged with `OriginTier` source authority (`PRIMARY_DOCUMENT` Rank 100, `SECONDARY_ANNEX` Rank 50, `EXTERNAL_INJECTED` Rank 10), section locators, and numerical/date tokens, alongside the deterministic cross-document clause diff matrix (`compute_clause_diffs`). The LLM has three tools:
    1. `get_evidence_overview()` (returns `OriginTier` hierarchy inventory, redline diff pairs, and controlling playbooks/directives/tables)
    2. `get_clause_diffs(filter_query, offset, limit)` (queries the deterministic clause-by-clause redline diff matrix and quantitative deltas)
    3. `query_claims_and_provisions(query, filename_filter, top_k)` (queries typed claim units with authority rank and section/quantitative metadata)

2.2 Study A Paired Results (Same Solver & Same Judge Across All 4 Tasks)

Task (`practice-area / slug`)	Docs	Criteria	`direct_rag` Passed	`direct_claimlayer` Passed	Difference (pp)	Criterion Transitions (Gain / Loss)	Oracle Union
`litigation-dispute-resolution` `analyze-counterparty-motion-to-dismiss`	6	34	26 / 34 (76.47%)	**27 / 34 (79.41%)**	**+2.94**	+3 / −2	29 / 34 (85.29%)
`structured-finance-securitization` `compare-offering-memorandum-against-indenture`	3	37	11 / 37 (29.73%)	**11 / 37 (29.73%)**	**+0.00**	+0 / −0	11 / 37 (29.73%)
`corporate-ma` `analyze-counterparty-spa-markup`	6	62	13 / 62 (20.97%)	**32 / 62 (51.61%)**	**+30.65**	**+23 / −4**	36 / 62 (58.06%)
`banking-finance` `analyze-counterparty-markup-of-senior-secured-credit-`	5	66	23 / 66 (34.85%)	**53 / 66 (80.30%)**	**+45.45**	**+30 / −0**	53 / 66 (80.30%)

Task (`practice-area / slug`)	Docs	Criteria	`direct_rag` Passed	`direct_claimlayer` Passed	Difference (pp)	Criterion Transitions (Gain / Loss)	Oracle Union
`facility-term-sheet`							
Total (Pooled, Single Judge)	**20**	**199**	**`73 / 199` (`36.68%`)**	**`123 / 199` (`61.81%`)**	**`+25.13`**	**`+56 / −6`**	**`129 / 199` (`64.82%`)**

2.3 Study A Summary Metrics & Resource Trade-offs

Measure	`direct_rag`	`direct_claimlayer`	Difference / Ratio
Pooled Criteria Passed	`73 / 199` (`36.68%`)	`123 / 199` (`61.81%`)	**`+50` criteria (`+25.13 pp`, `1.68x`)**
Unweighted Mean of Task Pass Rates	`40.51%`	`60.26%`	**`+19.75 pp`**
Criterion-Level Gains / Losses	—	`+56` gained / `−6` lost	Net `+50` criteria
Oracle Criterion Union (Post-hoc across both arms)	—	`129 / 199` (`64.82%`)	`+6` criteria above `direct_claimlayer` alone
Tool Calls Executed per Task	`5` (`20` total)	`5` (`20` total)	Matched (`5` tool calls + `1` synthesis call)
Recorded Solver Tokens	`142,756`	`640,546`	`+497,790` tokens (`4.49x` context volume)

2.4 Key Takeaways from Study A

1. **Substantial Gain on Redline & Quantitative Contract Tasks (`+30.65 pp` and `+45.45 pp`):** When the LLM is restricted to 5 tool-calling rounds without `bash`, standard BM25 chunk search (`direct_rag`) misses scattered clause modifications and fails to align counterparty markups with governing playbooks and financial tables (`13/62` on Corporate M&A; `23/66` on Banking & Finance). Providing structured claim units, `OriginTier` authority inventory, controlling tables, and deterministic clause-by-clause diffs (`direct_claimlayer`) raises Corporate M&A to **`32/62` (`51.61%`, `+23 / −4`)** and Banking & Finance to **`53/66` (`80.30%`, `+30 / −0`)** in a single rollout ($k = 1$).
2. **Elimination of the Banking Regression Seen in Study B (`+30 / −0` vs. `+13 / −12`):** Removing the 80-paragraph indexing cap (`paragraphs[:80]`) present in the legacy `deltoryx_audit.py` script and exposing both the controlling playbook/directives and the full-text redline diff matrix through structured tools eliminated the 12 criterion losses observed in the `OpenCode` agent run.
3. **Context Density vs. Token Expenditure:** Because `get_evidence_overview` and `get_clause_diffs` return rich structured tables and redline pairs per call, `direct_claimlayer` consumes `640,546` solver tokens across the 4 tasks versus `142,756` for `direct_rag`.

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3. Study B: Exploratory Autonomous Agent Harness Integration (`OpenCode` / `Harness²`)

3.1 Paired Task-Level Breakdown (After Container Repair, Same Judge Within Each Task)

Task (practice-area / slug)	Docs	Criteria	Base Agent Passed	Augmented Agent Passed	Difference (pp)	Criterion Transitions (Gain / Loss)	Judge
litigation-dispute-resolution analyze-counterparty-motion-to-dismiss	6	34	1 / 34 (2.94%)	**19 / 34 (55.88%)**	**+52.94**	+18 / −0	openai/gpt-4.1
structured-finance-securitization compare-offering-memorandum-against-indenture	3	37	3 / 37 (8.11%)	**12 / 37 (32.43%)**	**+24.32**	+10 / −1	nvidia/nemotron-3-super-120b-a12b
corporate-ma analyze-counterparty-spa-markup	6	62	15 / 62 (24.19%)	**24 / 62 (38.71%)**	**+14.52**	+12 / −3	nvidia/nemotron-3-super-120b-a12b
banking-finance analyze-counterparty-markup-of-senior-secured-credit-facility-term-sheet	5	66	43 / 66 (65.15%)	**44 / 66 (66.67%)**	**+1.52**	+13 / −12	openai/gpt-4.1
Total (Pooled)	**20**	**199**	**62 / 199 (31.16%)**	**99 / 199 (49.75%)**	**+18.59**	**+53 / −16**	**Mixed (Matched per task)**

3.2 Methodological Notes on Study B

- **Container Repair & Judge Pairing:** An earlier draft reported 47/199 vs. 106/199, which combined a 31/62 GPT-4.1 evaluation on the repaired augmented Corporate M&A artifact with a 0/62 pre-repair base M&A score (when both M&A arms scored 0/62 pre-repair due to raw Markdown written to .docx paths). Table 3.1 enforces matching judges within each task after container repair (15/62 vs. 24/62 under Nemotron).
- **Oracle Criterion Union vs. Whole-Deliverable Selection:** In Study B, the post-hoc criterion union across both arms is 115/199 (57.79%) and 56/66 (84.85%) in Banking. A whole-response selector is capped at 99/199 (49.75%).
- **Resource Use & Bundled Intervention:** Solver token usage in Study B rose from 220,320 (base) to 388,406 (augmented, +76.3%; logged solver cost USD 1.8000 vs. USD 2.4705, +37.3%), with always-on prompt characters increasing from 2,333 to 9,456.

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4. Repository Artifacts

- **Direct LLM Tool-Calling Runner (Study A):** benchmarks/run_direct_tool_benchmark.py
- **Direct LLM Tool-Calling Summary & Per-Task Scores (Study A):** benchmarks/direct_tool_results/direct_tool_summary.json
- **OpenCode Component Tree (Study B):** benchmarks/deltoryx_lab/
- **CLI Indexing & Verification Script (Study B):** benchmarks/deltoryx_lab/scripts/deltoryx_audit.py
- **OpenCode A/B Benchmark Runner (Study B):** benchmarks/run_ab_benchmark.py